Topic 3 — Electric Circuits

3.1 Charge, Current, and Potential Difference

Electric current (I): The rate of flow of charge past a point in a circuit. Units: A (amperes).
Charge (Q): Measured in coulombs (C). One coulomb = the charge flowing when 1 A passes for 1 s.
Potential difference (V): The energy transferred per unit charge between two points. Units: V (volts). 1 V = 1 J C^{-1}.

$$Q = It$$

$$V = W / Q$$

Conventional current flows from positive to negative (the direction positive charges would move). In metals, the actual charge carriers are free (delocalised) electrons moving from negative to positive. In electrolytes, both positive and negative ions carry charge.

The Drift Velocity Equation

The current in a conductor depends on the number density of charge carriers (n), their charge (q), the cross-sectional area (A), and their mean drift velocity (v):

$$I = nAqv$$

- n = number density of charge carriers (m^{-3}) — very high for metals (~10^{28}), much lower for semiconductors
- A = cross-sectional area (m^2)
- q = charge on each carrier (C) — for electrons, q = e = 1.60×10^{-19} C
- v = mean drift velocity (m s^{-1}) — typically very small (~10^{-4} m s^{-1} in metals)

Worked Example: A copper wire of cross-sectional area 2.0×10^{-6} m^2 carries a current of 5.0 A. Copper has $n = 8.5 \times 10^{28}$ m^{-3}. Calculate the drift velocity.

Solution:
$v = I/(nAq) = 5.0 / (8.5 \times 10^{28} \times 2.0 \times 10^{-6} \times 1.60 \times 10^{-19})$
$= 5.0 / (2.72 \times 10^{4}) =$ **1.84×10^{-4} m s^{-1}** (about 0.18 mm/s)

Exam Tip: The drift velocity is extremely slow, but the electric signal travels at close to the speed of light. This is because the electric field propagates almost instantaneously through the wire, causing all electrons to start drifting at the same time — like a pipe full of water where pushing one end causes water to flow out the other end immediately.

3.2 Resistance and Ohm's Law

Resistance (R): The opposition to current flow through a component. R = V/I. Units: Ω (ohms). 1 Ω = 1 V A^{-1}.

$$V = IR \quad (\text{or equivalently } R = V / I)$$

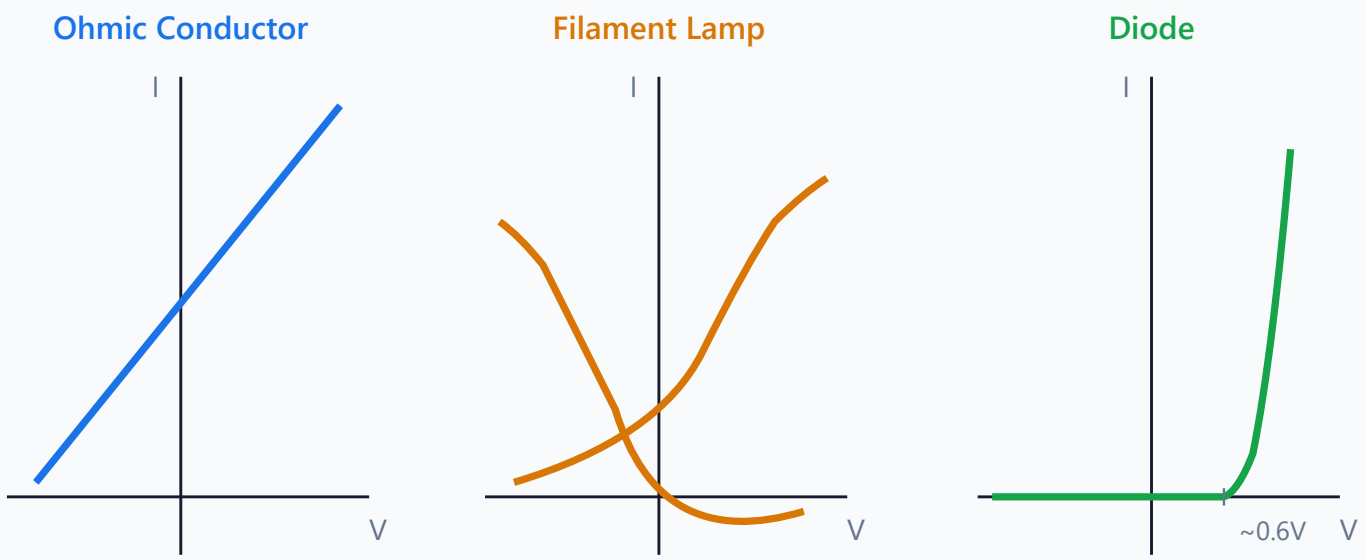
Ohm's Law: For a conductor at constant temperature, the current through it is directly proportional to the potential difference across it. Such a component is called an **ohmic conductor** and has constant resistance.

I-V Characteristics

Component	I-V Graph Shape	Key Features
Ohmic conductor (wire at constant T)	Straight line through origin	Resistance is constant; I is proportional to V. Gradient = 1/R.

Filament lamp	S-shaped curve through origin (less steep at higher V)	Resistance increases as temperature rises; higher current heats the filament, increasing lattice vibrations and thus resistance.
Diode	Flat for reverse bias; steep exponential rise above ~0.6 V forward	Very high resistance in reverse. Low resistance above threshold voltage (~0.6 V for silicon).
NTC Thermistor	(R decreases as T increases)	At higher temperatures, more charge carriers are freed from atoms, increasing n. The decrease in resistance due to increased n outweighs the increase due to more collisions.
LDR	(R decreases as light increases)	Photons free more charge carriers from the semiconductor material, increasing n and reducing resistance.

I-V Characteristic Curves



Exam Tip: For I-V characteristics, the gradient of the graph is NOT resistance (unless it passes through the origin as a straight line). Resistance at any point = V/I (the ratio), not ΔV/ΔI (the gradient). For a filament lamp, the resistance at any point is found by drawing a line from the origin to that point and calculating V/I.

3.3 Resistivity

Resistivity (ρ): A material property that measures how strongly a material opposes current flow, independent of the sample's dimensions. Units: Ω m.

$$\rho = RA / L$$

Rearranging: $R = \rho L/A$. This shows that resistance is proportional to length and inversely proportional to cross-sectional area.

Factors Affecting Resistance

- R increases with length ($R \propto L$) — more material for electrons to pass through
- R decreases with cross-sectional area ($R \propto 1/A$) — more paths available for current
- R depends on the material (through ρ) — copper has low ρ, rubber has high ρ
- For metals, R increases with temperature — more lattice vibrations impede electron flow
- For semiconductors (NTC thermistors), R decreases with temperature — more charge carriers freed

Worked Example: A constantan wire has a diameter of 0.36 mm and a length of 0.80 m. Its resistance is measured as 5.4 Ω. Calculate the resistivity of constantan.

Solution:

$d = 0.36 \times 10^{-3}$ m, so $r = 0.18 \times 10^{-3}$ m

$A = \pi r^2 = \pi \times (0.18 \times 10^{-3})^2 = 1.018 \times 10^{-7}$ m^2

$\rho = RA/L = (5.4 \times 1.018 \times 10^{-7}) / 0.80 =$ **6.9×10^{-7} Ω m**

(The accepted value is ~4.9×10^{-7} Ω m; the discrepancy could be due to contact resistance or diameter measurement error.)

Exam Tip: Remember to convert diameter to metres, then find radius, then use $A = \pi r^2$. A very common error is forgetting to halve the diameter to get the radius. Also: the percentage uncertainty in area is **twice** the percentage uncertainty in diameter, because $A = \pi(d/2)^2$ means $A \propto d^2$.

Superconductors

Superconductor: A material whose resistance drops to exactly zero below a critical temperature (transition temperature). Current flows indefinitely without energy loss.

Applications: MRI scanners, particle accelerators (e.g. LHC), maglev trains, lossless power transmission. The main limitation is the requirement for very low temperatures (often requiring expensive liquid helium cooling at ~4 K).

3.4 EMF and Internal Resistance

Electromotive force (EMF, ε): The total energy transferred per unit charge by the power supply. It is the PD across the terminals when no current flows (open circuit). Units: V. EMF is not a force.

Internal resistance (r): The resistance within the power supply itself. Energy is dissipated inside the source as thermal energy, reducing the terminal PD.

$$\varepsilon = I(R + r) = V + Ir$$

Rearranging: **V = ε − Ir**. The terminal PD (V) decreases linearly as current (I) increases.

Lost Volts

The **lost volts** (v_{lost} = Ir) is the potential difference across the internal resistance. It represents the energy per coulomb wasted as heat inside the battery. As current increases, more energy is wasted internally and less is available to the external circuit.

Worked Example: A cell has an EMF of 1.50 V and internal resistance 0.80 Ω. It is connected to a 4.0 Ω external resistor. Calculate (a) the current, (b) the terminal PD, (c) the lost volts, (d) the power dissipated internally.

Solution:
(a) I = ε/(R + r) = 1.50/(4.0 + 0.80) = 1.50/4.80 = **0.3125 A**
(b) V = IR = 0.3125 × 4.0 = **1.25 V**

(c) Lost volts = Ir = 0.3125 × 0.80 = **0.25 V**. Check: V + Ir = 1.25 + 0.25 = 1.50 V = ε. Correct.

(d) $P_{internal} = I^2r = (0.3125)^2 \times 0.80 =$ **0.078 W**

Graphical Determination of ε and r

Plot V (y-axis) vs I (x-axis). From V = ε − Ir:

- Y-intercept = ε (EMF)
- Gradient = −r (so r = magnitude of gradient)
- X-intercept = ε/r (short-circuit current)

3.5 Kirchhoff's Laws

Kirchhoff's First Law (Junction Rule): The sum of currents entering a junction equals the sum leaving. Based on conservation of charge.

$\Sigma I_{in} = \Sigma I_{out}$

Kirchhoff's Second Law (Loop Rule): The sum of EMFs around any closed loop equals the sum of PDs around that loop. Based on conservation of energy.

$\Sigma\varepsilon = \Sigma IR$

Worked Example: At a junction, three wires meet. Wire 1 carries 2.0 A into the junction. Wire 2 carries 0.5 A out. What is the current in wire 3, and in which direction?

Solution:

By K1L: $I_{in} = I_{out}$

$2.0 = 0.5 + I_3$

I_3 = **1.5 A out of the junction**

3.6 Series and Parallel Circuits

Series Circuits

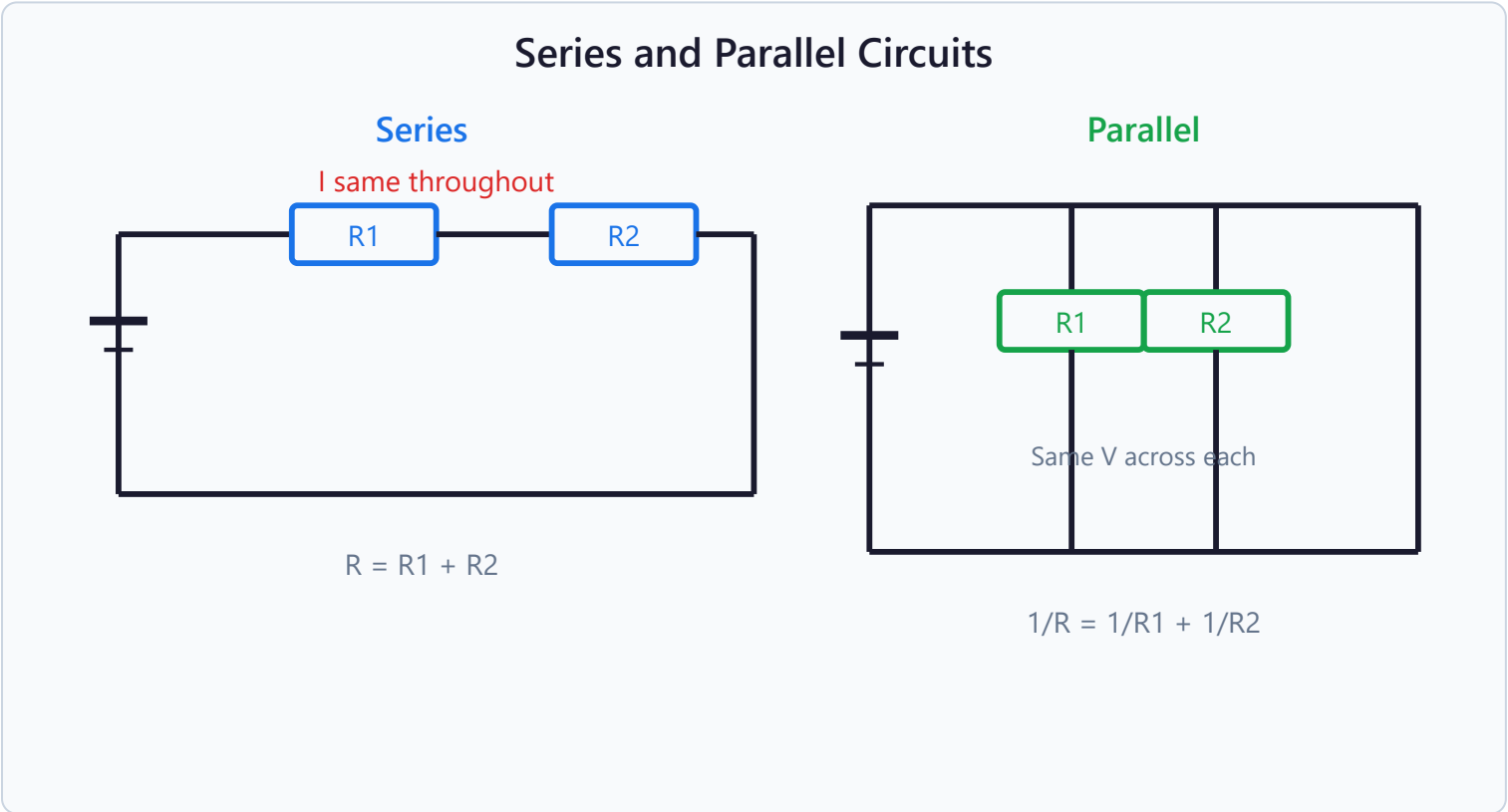
- Current is the same through all components: $I_{total} = I_1 = I_2 = I_3$
- Total PD is shared: $V_{total} = V_1 + V_2 + V_3$
- Total resistance: $R_{total} = R_1 + R_2 + R_3$ (always larger than the biggest R)

Parallel Circuits

- PD is the same across each branch: $V_{total} = V_1 = V_2 = V_3$
- Total current splits between branches: $I_{total} = I_1 + I_2 + I_3$
- Combined resistance: $1/R_{total} = 1/R_1 + 1/R_2 + 1/R_3$ (always smaller than the smallest R)

Shortcut for Two Resistors in Parallel

$$R_{total} = (R_1 \times R_2) / (R_1 + R_2)$$



Worked Example: A 12 Ω and a 6.0 Ω resistor are connected in parallel. This combination is in series with a 4.0 Ω resistor and a 6.0 V battery of negligible internal resistance. Find (a) the total resistance,

(b) the current from the battery, (c) the PD across the parallel combination.

Solution:

(a) Parallel: $R_p = (12 \times 6)/(12 + 6) = 72/18 = 4.0\ \Omega$

Total: $R_T = 4.0 + 4.0 =$ **8.0 Ω**

(b) $I = V/R = 6.0/8.0 =$ **0.75 A**

(c) $V_{parallel} = IR_p = 0.75 \times 4.0 =$ **3.0 V**

Exam Tip: The combined resistance of parallel resistors is always LESS than the smallest individual resistor. A quick check: if you get a parallel resistance larger than the smallest resistor, you have made an error.

3.7 Potential Dividers

A **potential divider** is two (or more) resistors in series across a supply voltage, producing an output voltage that is a fraction of the input.

$$V_{out} = V_{in} \times R_2 / (R_1 + R_2)$$

where V_{out} is measured across R_2. The output voltage is proportional to R_2's share of the total resistance.

Sensor Circuits Using Potential Dividers

Sensor	Condition Change	V_{out} across sensor	V_{out} across fixed R
NTC Thermistor	Temperature rises	Decreases	Increases
LDR	Light increases	Decreases	Increases

By choosing which resistor V_{out} is taken across, you can make the output voltage increase or decrease with temperature/light as needed for a particular application.

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Worked Example: A thermistor (R = 2.0 kΩ at 20°C, R = 0.50 kΩ at 80°C) is in series with a 1.0 kΩ fixed resistor across a 5.0 V supply. V_{out} is taken across the fixed resistor. Find V_{out} at 20°C and 80°C.

Solution:
At 20°C: V_{out} = 5.0 × 1.0/(2.0 + 1.0) = 5.0/3.0 = **1.67 V**
At 80°C: V_{out} = 5.0 × 1.0/(0.50 + 1.0) = 5.0/1.5 = **3.33 V**
V_{out} increases with temperature (across the fixed resistor), which could trigger a fan or alarm at a set threshold.

Exam Tip: The potential divider equation only applies when no current is drawn from the output (unloaded). If a component is connected across V_{out}, it effectively acts in parallel with R_2, reducing the effective resistance and lowering V_{out}. This is called the **loading effect**.

3.8 Energy and Power in Circuits

Electrical power (P): The rate of electrical energy transfer. Units: W (watts).

$$P = IV = I^2R = V^2/R$$

Energy Transferred

$$E = IVt = I^2Rt = V^2t/R$$

In a resistor, electrical energy is converted to thermal energy (heating). 1 kWh = 3.6×10^6 J.

Worked Example: An electric heater has a resistance of 20 Ω and is connected to a 230 V mains supply. Calculate (a) the current, (b) the power, (c) the cost of running it for 3 hours at 15p per kWh.

Solution:
(a) I = V/R = 230/20 = **11.5 A**

(b) $P = V^2/R = 230^2/20 = 52900/20 =$ **2645 W** ≈ 2.65 kW

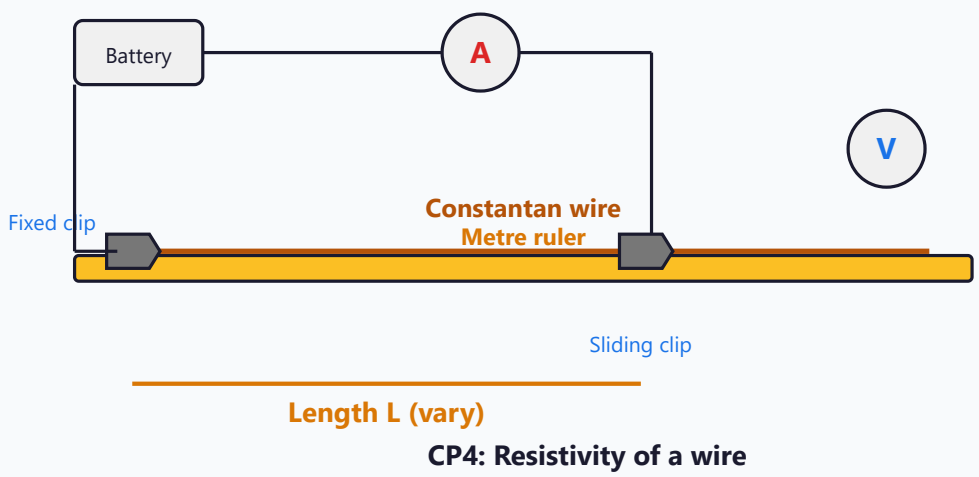
(c) Energy = 2.645 kW × 3 h = 7.935 kWh. Cost = 7.935 × 15p = **119p** ≈ £1.19

Exam Tip: Use $P = I^2R$ when you know I and R. Use $P = V^2/R$ when you know V and R. Power dissipated in internal resistance is "wasted" — it heats the battery. For maximum power transfer to the load, set R = r (load matches internal resistance).

Summary of Key Equations

Quantity	Equation
Charge and current	$Q = It$; $I = nAqv$
Resistance	$R = V/I$; $\rho = RA/L$
EMF and internal resistance	$\varepsilon = V + Ir = I(R + r)$
Series resistors	$R_T = R_1 + R_2 + \ldots$
Parallel resistors	$1/R_T = 1/R_1 + 1/R_2 + \ldots$
Potential divider	$V_{out} = V_{in} R_2/(R_1 + R_2)$
Power	$P = IV = I^2R = V^2/R$
Energy	$E = IVt = Pt$

⚗ Core Practical 4: Determine the resistivity of a wire



CP4: Resistivity of a wire

Aim

To determine the resistivity ρ of constantan wire by measuring how R varies with L.

Apparatus

Constantan wire (SWG 28 or 32, ~1 m), metre ruler, micrometer screw gauge (0.01 mm), two crocodile clips, ammeter, voltmeter, battery/power supply (low voltage ~1.5 V), switch, connecting wires.

Method

1. Measure the wire diameter d with the micrometer at 3+ positions, rotating 90 degrees between readings. Calculate mean d and area $A = \pi(d/2)^2$.
2. Tape the wire along the ruler. Fix one clip at 0 cm. Connect ammeter in series, voltmeter in parallel across the wire section.
3. Set the sliding clip to L = 20 cm. Close the switch briefly, read V and I, switch off. Calculate R = V/I.
4. Repeat for L = 30, 40, 50, 60, 70, 80, 90, 100 cm.
5. Plot R vs L. Gradient = ρ/A. Therefore ρ = gradient × A.

Safety

Use low voltage (≤ 1.5 V). Switch off between readings to prevent heating. Do not touch wire while current flows.

Sources of Error & Improvements

Source of Error	Type	Improvement
Wire heats up, changing resistance	Systematic	Low current; switch off between readings
Diameter varies along wire	Random	Measure at 6+ points and orientations; use mean
Zero error on micrometer	Systematic	Check zero reading; subtract from all measurements
Contact resistance at clips	Systematic	Clean clips; firm contact; y-intercept on graph shows this

Exam Tip: The biggest uncertainty is usually in the diameter because it is squared in the area calculation. A 5% error in d gives ~10% error in A and hence ρ. Constantan is used because its resistivity barely changes with temperature.

⚗ Core Practical 5: Determine EMF and internal resistance of a cell

Aim

To determine the EMF (ε) and internal resistance (r) of a cell by plotting terminal PD against current.

Apparatus

Cell (1.5 V AA), variable resistor (rheostat, 0-10 Ω), ammeter (0-1 A), voltmeter (high resistance, 0-2 V), switch, connecting wires.

Method

1. Connect cell, switch, ammeter, and variable resistor in series. Voltmeter in parallel across the cell terminals.
2. Close switch, adjust rheostat to maximum R. Record V and I.
3. Decrease R in steps, recording V and I each time (at least 6 pairs).
4. Open switch between readings to prevent cell running down.
5. Plot V vs I. Y-intercept = ε. Gradient = −r.

Safety

Never short-circuit the cell (R = 0). Switch off between readings. Handle cells carefully.

Sources of Error & Improvements

Source of Error	Type	Improvement
Cell runs down during experiment	Systematic	Switch off between readings; use fresh cell; take readings quickly
Cell heats up, changing r	Systematic	Allow cooling; use moderate currents
Voltmeter draws small current	Systematic	Use high-resistance digital voltmeter
Contact resistance	Random	Firm, clean connections

Exam Tip: The y-intercept gives ε more accurately than a single open-circuit reading because it uses all data points. The magnitude of the gradient gives r. A steeper negative slope means higher internal resistance.

⚗ Core Practical 13: Calibrate a thermistor in a potential divider as a thermostat

Aim

To calibrate an NTC thermistor in a potential divider circuit by plotting V_{out} against temperature, producing a calibration curve that can be used as an electronic thermometer or thermostat.

Apparatus

NTC thermistor, fixed resistor (value similar to thermistor at mid-range T), DC power supply (5 V), voltmeter (across thermistor), thermometer (0-100°C), beaker of water, heater/Bunsen, stirrer, connecting wires.

Method

1. Set up the potential divider: supply in series with R_1 (fixed) and thermistor. Voltmeter across the thermistor.
2. Place the thermistor (waterproofed) in the water bath with the thermometer.
3. Record temperature θ and V_{out} at room temperature.
4. Heat slowly, stirring continuously. Record V_{out} every 5°C from 20°C to 80°C.
5. Allow thermal equilibrium at each temperature before reading.
6. Plot the calibration curve: V_{out} vs θ.

Analysis

$V_{out} = V_s \times R_{thermistor}/(R_1 + R_{thermistor})$. As T increases, $R_{thermistor}$ decreases (NTC), so V_{out} decreases. The calibration curve allows conversion of any measured V_{out} to a temperature.

Safety

Take care with hot water (≤ 80°C). Keep electrical connections away from water. Use a heatproof mat.

Sources of Error & Improvements

Source of Error	Type	Improvement
Temperature not uniform in bath	Random	Stir continuously; wait for equilibrium

Thermistor not at same T as thermometer	Systematic	Place both sensors close together; allow time
Self-heating of thermistor by current	Systematic	Use low supply voltage

Exam Tip: Remember: when V_{out} is measured across the thermistor, it decreases as temperature rises (because $R_{thermistor}$ falls). If the thermistor and fixed resistor swap positions, V_{out} increases with temperature instead. Always check which component V_{out} is across.